

## Lesson 7 Biconditionals; Proof by Contrapositive

In Lesson 6 we highlighted the difference between proving (and disproving) universal and existence statements. We also found that many theorems are conditionals. In this lesson, we prove theorems which are **biconditionals** and introduce an indirect technique to prove conditionals: **proof by contrapositive**.

Since  $p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$ , to prove a biconditional, we prove both directions, i.e. two conditionals:  $p \rightarrow q$  and its converse  $q \rightarrow p$ .

**Proposition 1** For every integer  $x$ ,  $x$  is odd if and only if  $x - 1$  is even.

**Proof** ( $\Rightarrow$ ) If  $x$  is odd, then  $x = 2n + 1$  for some  $n \in \mathbb{Z}$ , and  $x - 1 = 2n$  is even.  
 ( $\Leftarrow$ ) Conversely, if  $x - 1$  is even, then  $x - 1 = 2n$  for some integer  $n$ , and  $x = 2n + 1$  is odd.

In this case, we may prove both directions *at once* as we only use definitions and equalities, which are bidirectional, *but* proving each direction separately is clearer.

**Proof 2** For any integer  $x$ ,  $x$  is odd if and only if  $x = 2n + 1$  for some integer  $n$ , which is true if and only if  $x - 1 = 2n$ . This is equivalent to  $x - 1$  is even.

Our proofs so far are **direct proofs**: we prove a conditional  $p \rightarrow q$  starting with the premise or hypothesis  $p$  and move forward to show that the conclusion  $q$  is true. We now introduce an indirect technique: **proof by contrapositive**.

- Instead of proving  $p \rightarrow q$  directly, we prove its contrapositive,  $\sim q \rightarrow \sim p$ , since a conditional is equivalent to its contrapositive.

**Proposition 2** For every integer  $x$ , if  $x^2$  is even, then  $x$  is even.

**Proof** We will prove the contrapositive of the proposition:

For every integer  $x$ , if  $x$  is odd, then  $x^2$  is odd.

Note that “not even” is “odd”, by the Theorem 3 of Lesson 6. If  $x$  is odd, then  $x = 2n + 1$  for some integer  $n$ , and

$$x^2 = (2n + 1)^2 = 4n^2 + 4n + 1 = 2(2n^2 + 2n) + 1$$

Since  $k = 2n^2 + 2n$  is an integer,  $x^2$  is odd.

The contrapositive is easier to prove, because it is easier to infer information about  $x^2$  from  $x$  than to get information about  $x$  from  $x^2$ . Since “ $x$  is even” means “ $2|x$ ”, we may not be surprised by a more general proposition,

For every integer  $x$  and any prime  $p$ , if  $p|x^2$ , then  $p|x$ .

The above proposition is an immediate *corollary* of the following theorem:

For all integers  $a, b$  and any prime  $p$ , if  $p|ab$ , then  $p|a$  or  $p|b$ . (\*)

If we *could* use the theorem (\*), then we didn't have to do the proof we did for Proposition 2! However, the proof of Proposition 2 is easier than the proof of (\*), which we will do in Lesson 10.

Sometimes it is equally easy to prove a conditional directly or its contrapositive.

**Proposition 3** Let  $a$  and  $b$  be real numbers. If  $ab = 1$ , then  $a \leq 1$  or  $b \leq 1$ .

**Proof 1** If  $a \leq 1$ , then we are done. If  $a > 1$ , then  $b = 1/a < 1$ , and  $b \leq 1$ .

**Proof 2** We prove the contrapositive: "If  $a > 1$  and  $b > 1$ , then  $ab \neq 1$ ."

If  $a > 1$  and  $b > 1$ , then  $ab > a(1) > a > 1$ , and  $ab \neq 1$ .

Before we do another example, we give a direct proof of a basic theorem.

**Proposition 4** The product of an even integer and any integer is even.

**Proof** Let  $x$  be an even integer and  $y$  be an integer. Then  $x = 2n$  for some integer  $n$ , and  $xy = (2n)y = 2(ny)$ . Since  $ny$  is an integer,  $xy$  is even.

**Proposition 5** If the product of two integers is odd, then the integers are not consecutive.

**Proof** We prove the contrapositive:

If two integers are consecutive, then their product is even.

Let  $x$  and  $x + 1$  be two consecutive integers. If  $x$  is even, then by Proposition 4,  $x(x + 1)$  is even. If  $x$  is odd, then  $x = 2n + 1$  for some integer  $n$ , and  $x + 1 = 2n + 2 = 2(n + 1)$  is even. So,  $x(x + 1)$  is even, again by Proposition 4.

We have seen that to prove a biconditional  $p \leftrightarrow q$ , we prove  $p \rightarrow q$  and  $q \rightarrow p$ . Sometimes, it may be easier to prove one direction using contrapositive. In other words, to prove  $p \leftrightarrow q$ , we may prove  $p \rightarrow q$  and its converse  $q \rightarrow p$ , or  $p \rightarrow q$  and its inverse  $\sim p \rightarrow \sim q$ .

**Proposition 6** For every integer  $x$ ,  $x$  is odd if and only if  $x^2 - 1$  is even.

**Proof** First, we prove that if  $x$  is odd, then  $x^2 - 1$  is even.

Let  $x$  be an odd integer, then  $x = 2n + 1$  for some integer  $n$ . So,

$$x^2 - 1 = (2n + 1)^2 - 1 = 4n^2 + 4n + 1 - 1 = 2(2n^2 + 2n)$$

Since  $2n^2 + 2n$  is an integer,  $x^2 - 1$  is even.

Next, we prove the inverse: if  $x$  is even (not odd), then  $x^2 - 1$  is odd (not even).

Let  $x$  be an even integer, then  $x = 2n$  for some integer  $n$ . So,

$$x^2 - 1 = (2n)^2 - 1 = 4n^2 - 1 = 4n^2 - 2 + 1 = 2(2n^2 - 1) + 1$$

Since  $2n^2 - 1$  is an integer,  $x^2 - 1$  is odd.

For this biconditional, it is easier to prove the inverse (if  $x$  is even, then  $x^2 - 1$  is odd) than the converse (if  $x^2 - 1$  is even, then  $x$  is odd).